Server



From Fragmentation to Standard: The OCP Footprint for AI Signal Conditioners

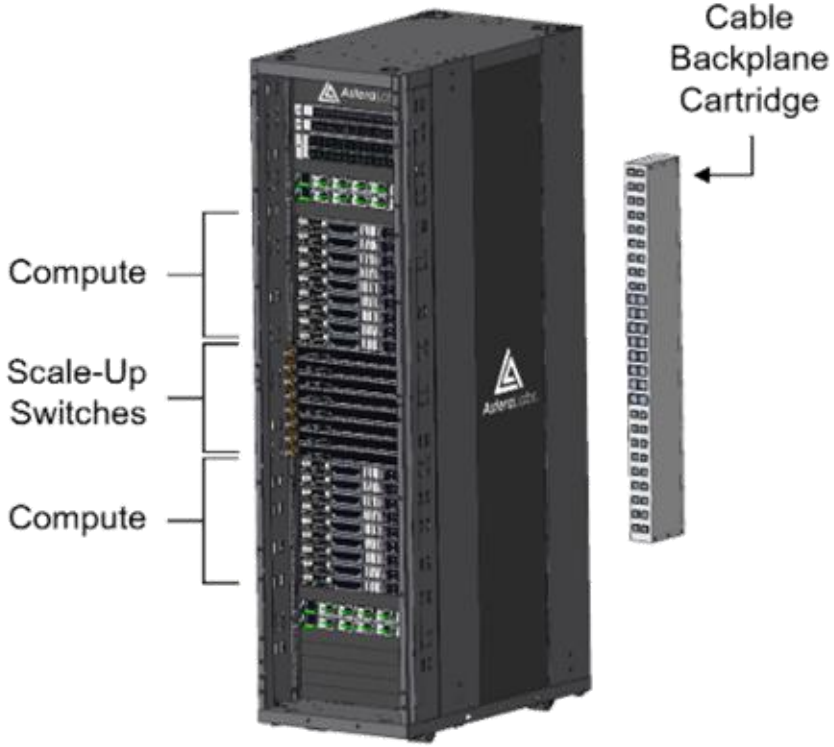
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OCP PLATINUM® Member

Challenges

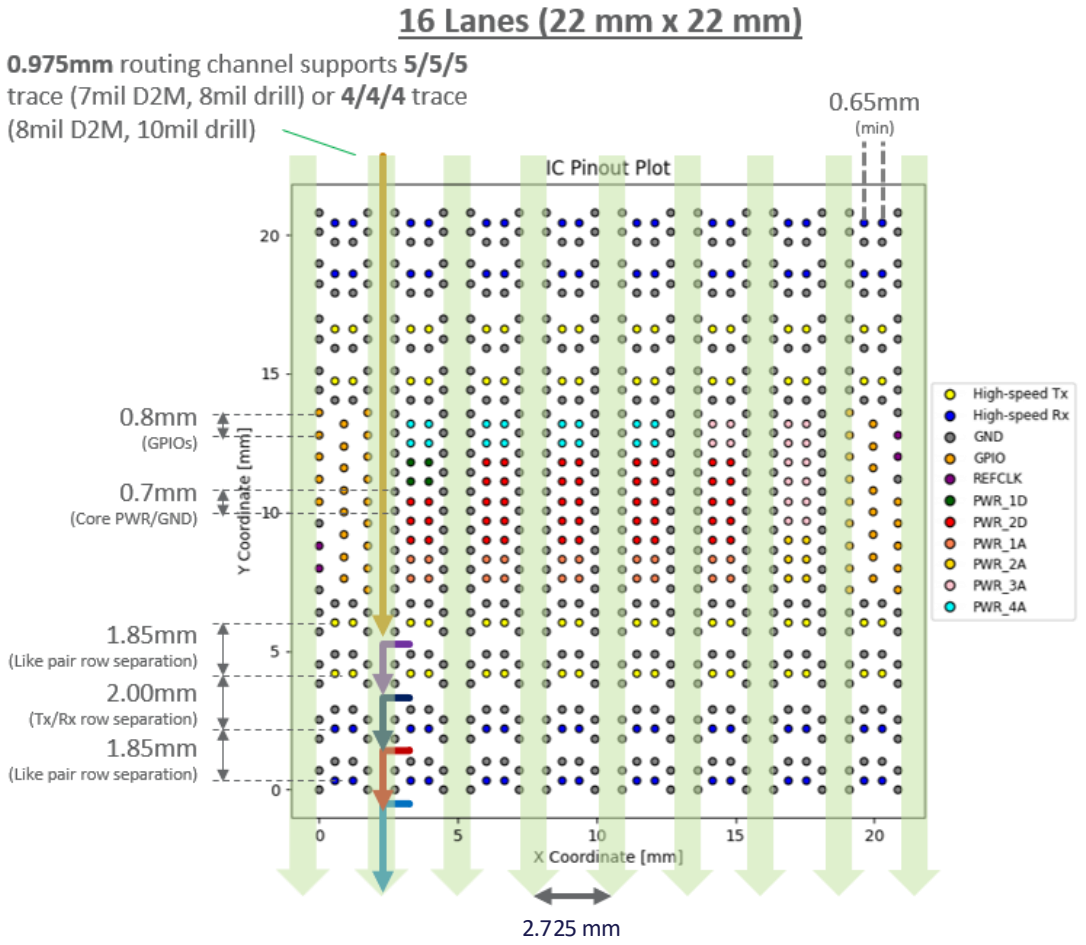
- AI scale-up backplanes **often require signal conditioners** (Retimers, Gearboxes, etc.) to support the high data rate and long reach
- Signal density is high, speeds are increasing, and PCB space is limited
- Conventional packages are **too large, lossy, and do not route cleanly** to mainstream backplane connectors
- Fine-pitch packages, while smaller, **do not support high-volume PCB design rules** for thick (e.g., 200 mil) compute and switch boards
- There is **no industry-standard pinout** for signal conditioner devices optimized for AI infrastructure; hard to design system boards for multiple vendors



OCP Signal Conditioner Standard Footprint

Key Features

- 8-Lane, 16-Lane, and 24-Lane variants
- Easily routable to mainstream backplane connectors (column pitches 2.4-2.8 mm)
- Scalable to support multiple row/column connector form factors
- Compact body size
- Conforms to DFM rules for thick PCBs (e.g., 200 mil)
  - Compatible with 10mil via drill size; optimized for 8mil drill
  - Compatible with 8mil drill-to-metal
- Routing channels thru pin field, no blocking vias, enables “tiling”
- Excellent FEXT/NEXT crosstalk characteristics with GND BGAs and vias surrounding all diff pairs
  - < -60 dB FEXT up to 54 GHz (200 mil stackup, 170 mil vias)
  - < -70 dB NEXT up to 54 GHz (200 mil stackup, 170 mil vias)
- Fully symmetric P/N breakout: zero skew



Scalable for Different Row/Column Connectors

"Tiling" multiple devices

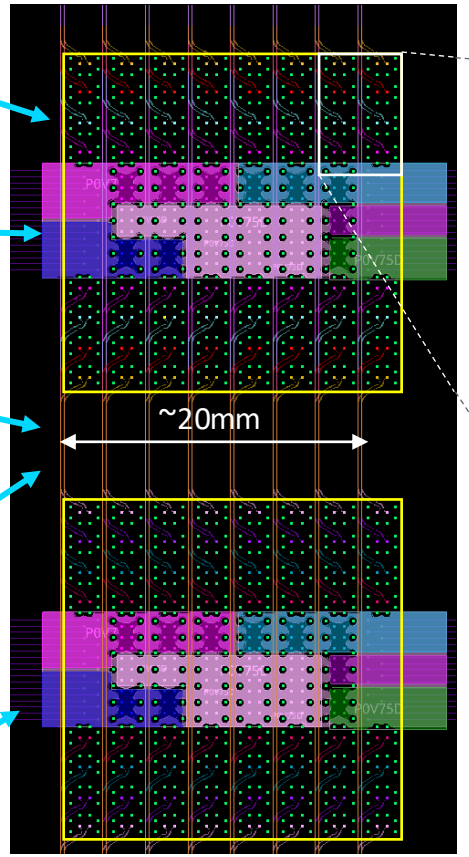
"**Tiling**" two Retimers to support 64 diff pairs in the ~22mm x-dimension

Power supply planes approach from sides

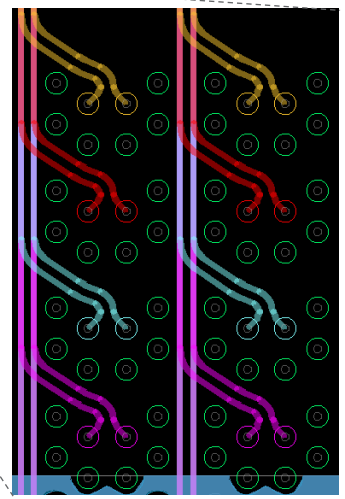
Flow-through routing

8 routing layers escape 64 diff pairs in a ~20-mm routing channel

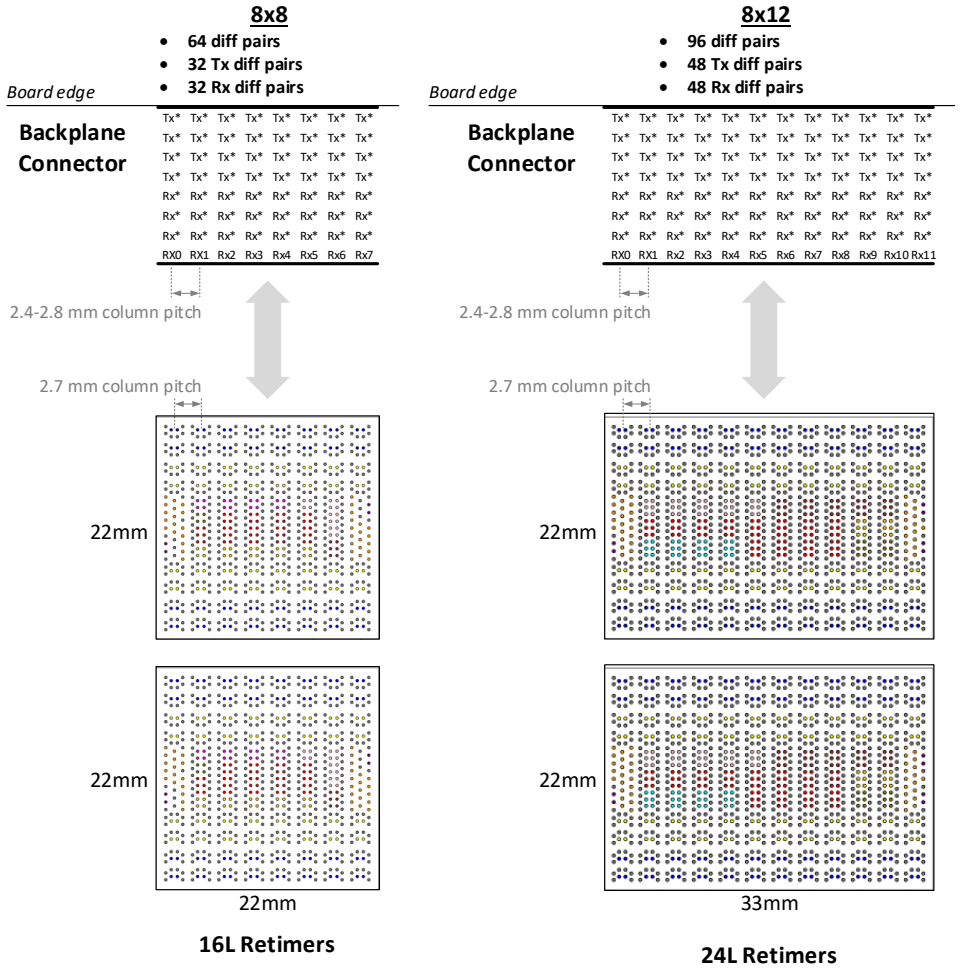
Second (or 3rd, 4th) device scales the signal density using same routing channels



Fully symmetric P/N breakout (zero skew)



Alignment to backplane connectors



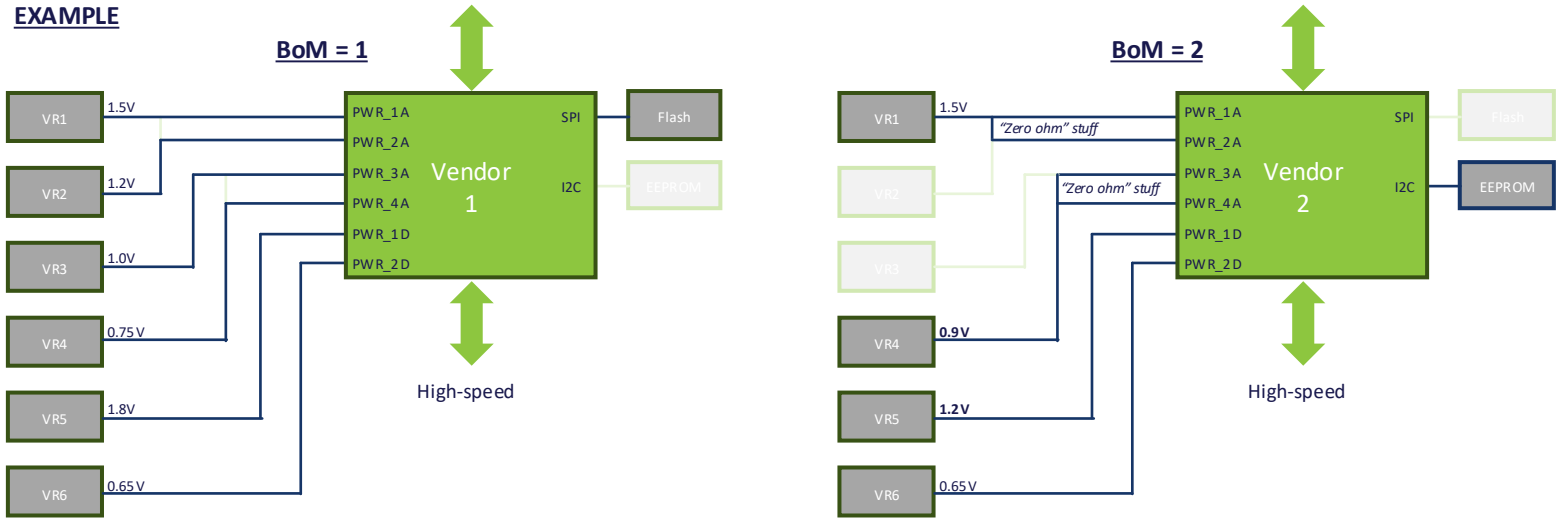
OCP Signal Conditioner Standard Footprint



	Redriver	Retimer	Gearbox
Function	Analog equalization	Clock and data recovery (CDR) and retransmission	Clock and data recovery (CDR), rate conversion, and retransmission
Application	Moderate signal reach extension (e.g., <20 dB)	Maximum signal reach extension (e.g., >20 dB)	Maximum signal reach extension and rate conversion (e.g., pairing current-gen switches with previous-gen GPUs)
Equalization	Linear, analog equalization only	Linear and non-linear equalization, digital and analog	Linear and non-linear equalization, digital and analog
Power	Lowest	Higher	Highest
Latency	Lowest	Higher	Highest

Multi-Vendor System Designs

- “BoM swap” compatibility is made possible by this spec
- Groupings of power supply pins facilitates “BoM swaps” – limits number of VRs and “zero ohm” resistors required to combine islands
- System designer must review datasheets from multiple vendors to determine whether “BoM swap” is indeed possible



Vendor 1 component: 6 supplies + SPI Flash

Vendor 2 component: 4 supplies + EEPROM

Summary

- A standard footprint optimized for **density, route-ability, and signal integrity** in AI infrastructure applications
- **Minimal body size** while **meeting HVM PCB design rules** for thick boards
- Future-proofed up to 128 GBaud (e.g., 256 Gbps PAM4)
- **Scalable** to support multiple row/column backplane connector form factors
- A design which is **signal conditioner technology agnostic** as well as **protocol agnostic**
- **“BoM swap compatibility”** between different device types and vendors

Call to Action

- v1.0.0 Base Specification was authored collaboratively between multiple OCP member companies
- As adoption picks up and system designers implement this footprint, feedback can be incorporated into future revisions of the Base Specification
- Future work:
  - Optimization for placement behind **I/O connectors** (OSFP, OSFP-XD, CDFP, etc.)
  - Optimization for **>128 GBaud/Lane**
  - Optimization for **CPC-to-Backplane interposers**

Download the Specification

Download the White Paper

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Thank You!

